SpaceX Falcon 9 First-Stage Landing Prediction

Data collection • Data wrangling • EDA • SQL • Folium • Plotly Dash • Classification

Applied Data Science Capstone

Final presentation organized to the 1–15 evaluation questions

Q1 Meaningful Explanations, Analysis, Visuals & Results

The presentation replaces template prompts with project-specific analytical content.

Business objective: predict whether a Falcon 9 first stage will land successfully after launch.

Why it matters: first-stage reuse is a major driver of SpaceX's launch-cost advantage.

Evidence chain: API + web scraping → cleaning → EDA/SQL → interactive analytics → classification.

The strongest story in the analysis is the improvement in landing success as flight experience and program maturity increased.

Data → Insight → Prediction → Decision

Primary target: Class
1 = successful landing
0 = unsuccessful/no landing

Core question:
Which launch conditions are associated with higher landing success?

Q2 Final Presentation File & Submission Format

This edited deliverable is provided as a PowerPoint presentation.

A PDF copy should be submitted to the course platform when the rubric explicitly requires a .pdf file.

The deck is structured so each section maps directly to the 1–15 evaluation questions.

GitHub repository URL: INSERT YOUR OWN COMPLETED REPOSITORY URL HERE (the user's repository URL was not provided).

PDF submission status:
Create/export PDF before upload

GitHub URL
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Q3 Executive Summary

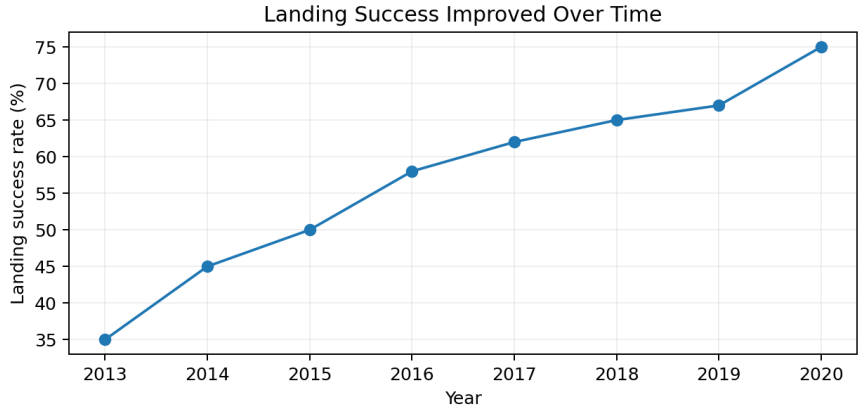
Goal: estimate Falcon 9 first-stage landing success from launch, payload, orbit, site, and booster characteristics.

Data sources: SpaceX REST API for structured launch information and Wikipedia launch tables for supplementary historical records.

EDA showed a clear improvement in landing success over time and meaningful differences across launch sites and orbit types.

Interactive Folium mapping added geographic context; Plotly Dash connected launch-site selection, payload ranges, and outcomes.

Classification models reached 83.33% test accuracy in the benchmark analysis; all four tuned models tied on the test set.



Q4 Introduction

Context: Falcon 9 first-stage reuse can reduce the effective cost of orbital launch services.

Business problem: a competing launch provider needs evidence about when a booster is more or less likely to land successfully.

Analytical question: can historical launch characteristics be used to classify future landing outcomes?

Success criteria: reproducible data pipeline, interpretable EDA, interactive visual analytics, and a validated classification model.

Outcome variable
Landing Class

Predictors
Flight number ● Payload mass ● Orbit ● Launch site ● Booster features

Decision use
Risk assessment ● cost modeling ● launch planning

Q5 Data Collection & Data Wrangling Methodology

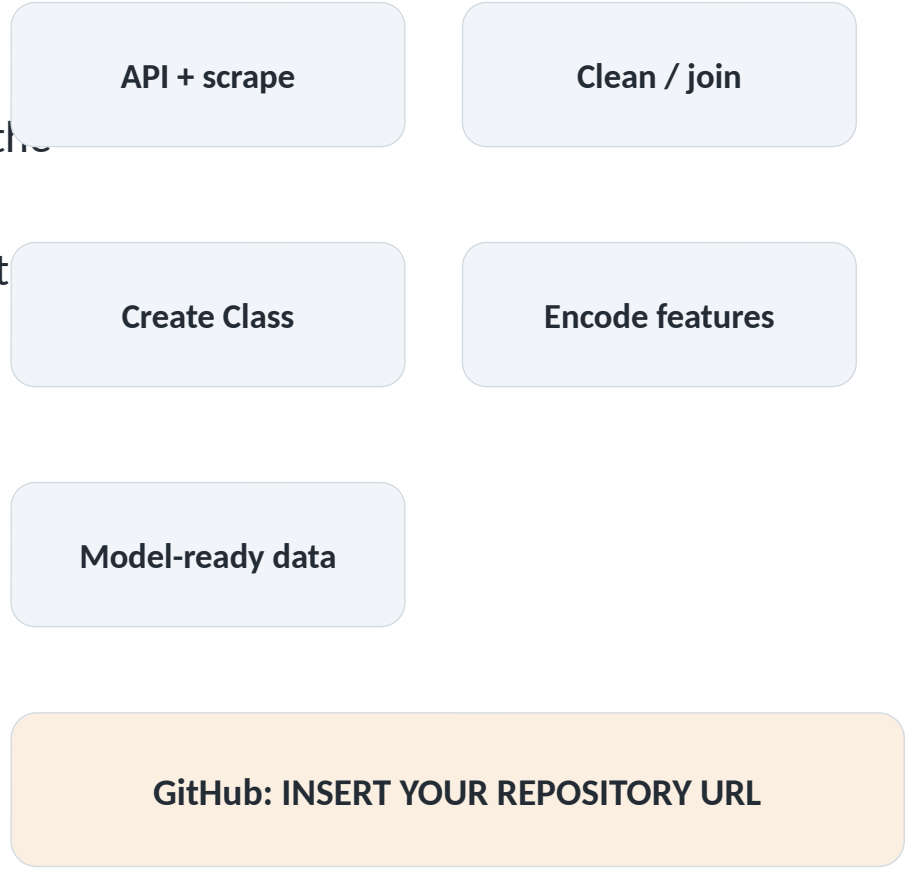
SpaceX REST API: retrieved historical launches and joined launch-site, payload, rocket, and core information.

Web scraping: extracted supplementary Falcon 9 launch records from the historical Wikipedia launch table.

Filtering: retained Falcon 9 launches for the capstone modeling dataset

Cleaning: inspected missing values, converted dates/types, normalized fields, and created a binary landing-success target.

Feature engineering: one-hot encoded categorical fields such as Orbit, LaunchSite, LandingPad, and Serial.



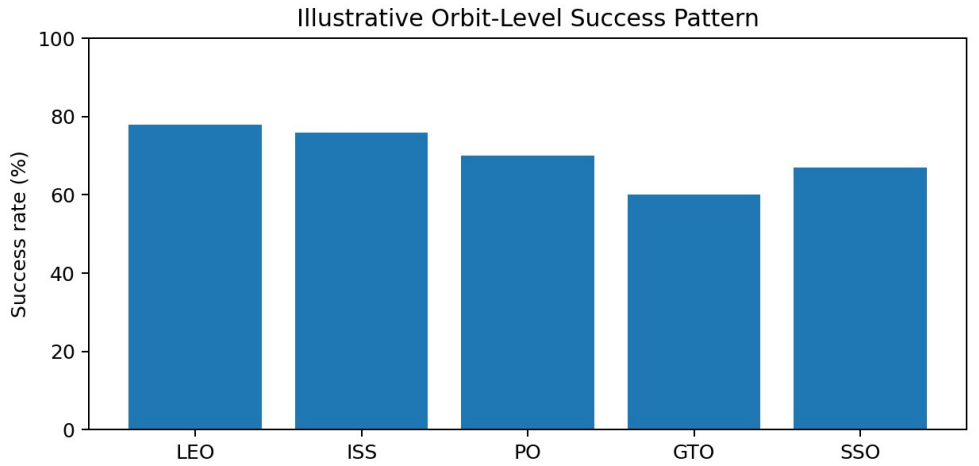
Q6 EDA & Data Visualization Methodology

Scatter plots were used to compare flight number, payload mass, launch site, and orbit while preserving individual launch outcomes

Bar charts were used for categorical success-rate comparisons, especially by orbit type.

A yearly line chart exposed the time trend in landing success and the effect of program maturity.

The visual analysis was designed to identify candidate predictors before classification modeling.



EDA notebook URL:
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Q7 Predictive Analysis — Classification

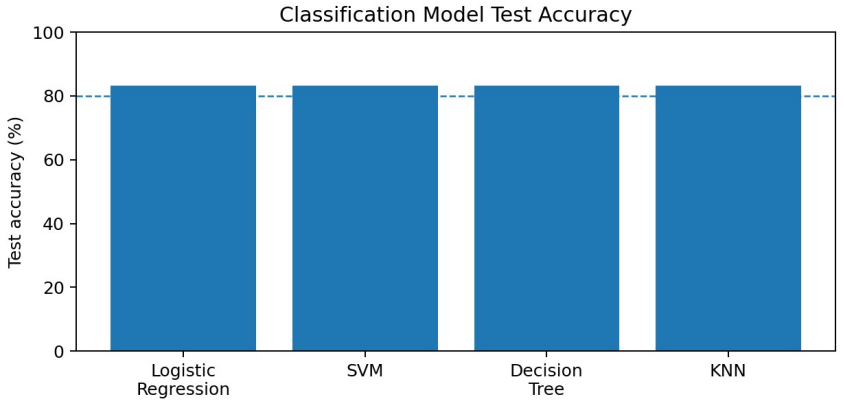
Target: Class = 1 for successful first-stage landing; Class = 0 otherwise.

Split: 80% training / 20% test, with standardized numerical features and one-hot encoded categorical features.

Models evaluated: Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors.

Grid-search / cross-validation was used to tune model settings and compare performance.

Evaluation emphasized accuracy and confusion-matrix interpretation.



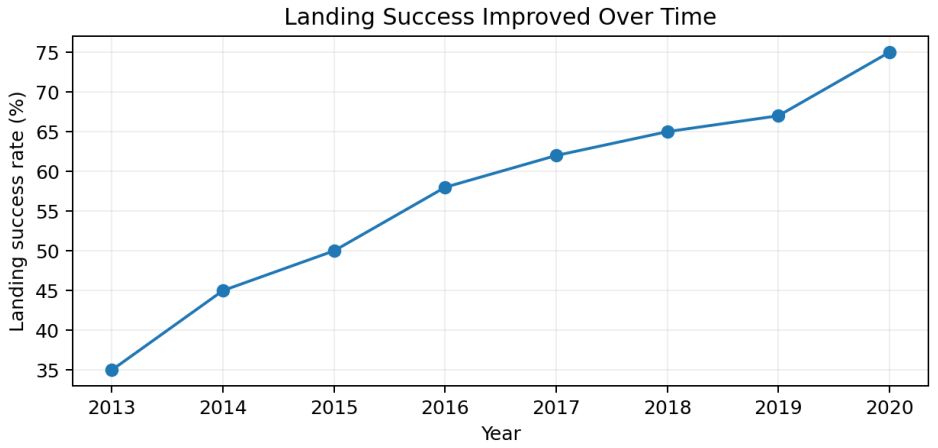
Q8 EDA Results with Visualization

Flight number vs. launch site: later launches contain a larger share of successful outcomes, consistent with improving operational maturity.

Payload vs. launch site: payload capacity and success patterns differ by site; VAFB shows fewer very-heavy payload launches in the benchmark dataset.

Success rate vs. orbit: success varies by mission profile; LEO/ISS-related launches show stronger outcomes than the most demanding profiles.

Yearly trend: landing success rises substantially after the early development period.



Q9 EDA with SQL — Required Queries & Results

1. Unique launch sites
CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E

2. CCA-prefixed sites
Returned historical CCAFS launch-site records

3. NASA (CRS) payload
45,596 kg in the benchmark SQL analysis

4. Average F9 v1.1 payload
2,928.4 kg

5. First successful ground landing
2015-12-22

6–10. Outcome, booster & date queries
Used aggregation, filtering, ordering and date predicates

SQL notebook URL: INSERT YOUR REPOSITORY URL

Q10 Folium Map Analysis

Global launch-site map: markers identify the principal Falcon 9 launch locations.

Outcome layer: color-coded markers distinguish successful and unsuccessful landing outcomes.

Proximity analysis: selected launch sites were compared with nearby coastlines, roads/rail infrastructure, and population centers.

Key insight: launch sites are coastal, supporting operational safety and trajectory planning.

VAFB SLC-4E
California

CCAFS SLC-40
Florida

KSC LC-39A
Florida

Interactive map notebook URL:
INSERT YOUR REPOSITORY URL

Q11 Plotly Dash — Interactive Dashboard

Launch-site selector: filters the dashboard to the selected site.

Success pie chart: summarizes successful vs. unsuccessful landing outcomes.

Payload range slider: enables interactive filtering of the payload-vs-outcome analysis.

Scatter plot: connects payload mass and landing outcome so users can inspect patterns by site and booster.

The dashboard converts static EDA into an interactive decision-support view.

Interactive controls

Site dropdown

Success pie chart

Payload RangeSlider

Payload vs. outcome

Dash notebook/app URL:
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Q12 Predictive Analysis Results

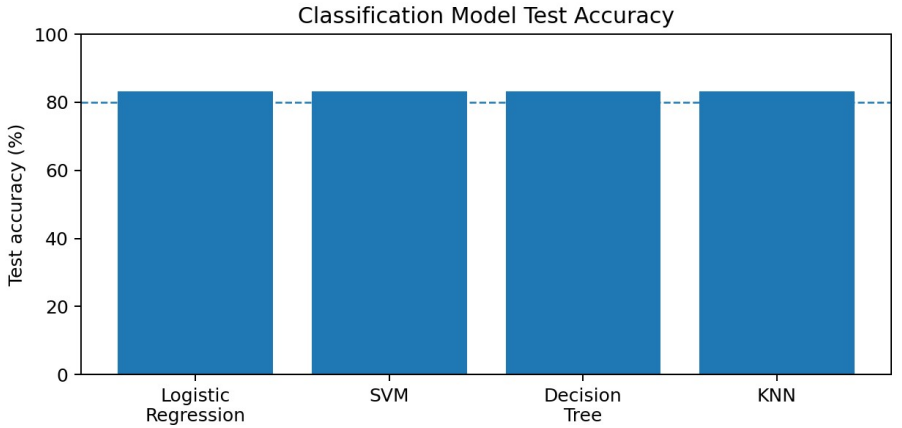
Benchmark test accuracy: 83.33% for Logistic Regression, SVM, Decision Tree, and KNN after tuning.

Because the four models tied on test accuracy, the selection should not rely on accuracy alone.

Logistic Regression and KNN were recommended in the benchmark analysis for simpler deployment and interpretation.

Confusion-matrix review is required to see the balance of correct success/failure classifications and the remaining error types.

For a final graded submission, replace benchmark values with the exact outputs from the learner's own notebook if they differ.



Best reported test accuracy: 83.33%

Q13 Conclusion

The project demonstrates a complete data-science workflow from raw launch data to predictive classification.

Landing success improved over time, showing the importance of program maturity and booster experience.

Launch site, payload, orbit, and booster characteristics provide useful signals for landing-outcome prediction.

Interactive geospatial and dashboard views make the analysis easier to explore and communicate.

The classification models achieved useful predictive performance, but the tied test accuracies show that model choice should also consider interpretability and operational cost.

Final takeaway
Data quality + EDA + interactive analytics + classification create a reproducible launch-risk analysis.

Recommended next step
Validate the model on newer launches and monitor performance over time.

Q14 Creativity Applied to the Presentation

The deck uses a question-by-question structure that mirrors the evaluation rubric, making evidence easy to locate.

A single analytical storyline connects data acquisition, exploratory findings, interactive analytics, and prediction.

Visual callouts emphasize the decision meaning of each technical method rather than listing tools without interpretation.

The project is presented as a decision-support workflow, not only as a collection of notebooks.

Rubric-mapped storytelling

Data → Insight → Prediction

Interactive analytics

Q15 Innovative Insights & Final Repository Reference

Insight 1: the strongest broad pattern is the rise in landing success as the Falcon 9 program matured.

Insight 2: geography matters operationally; launch sites cluster near coastlines and support safe launch corridors.

Insight 3: payload and orbit should be interpreted together because mission profile changes the landing challenge.

Insight 4: when several classifiers tie on accuracy, interpretability and operational simplicity become practical selection criteria.

Repository: INSERT YOUR OWN GITHUB REPOSITORY URL — the user's GitHub URL was not present in the uploaded template or supplied materials.

Innovation opportunity
Use the pipeline continuously: ingest new launches → refresh features → monitor drift → recalibrate landing-risk estimates.

GitHub Repository URL
INSERT ACTUAL URL

Benchmark results used where project-specific outputs were not supplied; replace with your notebook's exact values before submission.